

Autonomous Link Restoration and Capacity Management in Open Optical Networks for Connected Mobility Services

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Abstract *This demonstration integrates sensing with programmable packet-optical infrastructure. We showcase autonomous capacity upgrades and network restoration to maintain high-performance connectivity for demanding mobility applications when using real-world data from a sensorized vehicle platform prototype. ©2026 The Author(s)*

Overview

Network service lifecycle management — encompassing creation, updating, and termination — is handled by orchestration systems through well-defined autonomous functions, workflows, data models, and APIs [1]. These systems coordinate heterogeneous network and cloud resources to fulfill service demands in terms of capacity, latency, and computing requirements.

Network resources provide end-to-end connectivity between access terminals and cloud facilities, spanning heterogeneous segments and technologies — such as packet and optical switching — while guaranteeing both minimum data rate and maximum tolerated latency bounds. To this end, the orchestration system coordinates a set of technology-specific SDN controllers that interact with the underlying network devices through well-defined APIs — comprising specific protocols and data models — exposing an abstracted view of resource states, such as available optical spectrum and transceiver parameters, and enabling device programmability in terms of resource allocation and capability configuration to meet service demands. In this context, widely adopted control protocols include NETCONF, RESTful, and gNMI, complemented by standard data model definitions such as OpenConfig [2] and OpenROADM [3].

The network resource-state information collected and exposed by the SDN controllers is essential for the higher-level orchestration entity to trigger dynamic end-to-end path computation and resource selection algorithms (based on either heuristic or AI/ML approaches) upon the arrival of a new service demand. Concurrently, the orchestration system delegates to a cloud controller the selection and allocation of computing resources (CPUs, RAM, storage) required to host the applications underpinning those network services.

Fig. 1 depicts the targeted demonstration setup, comprising a hierarchical orchestration system that coordinates a set of technology-specific SDN controllers to autonomously manage connectivity services deployed across a disaggregated packet-

optical network. In the access domain, sensors and data sources/publishers (*Data pub.*) are distributed across both fixed (e.g., PON) and wireless terminals. The generated data traffic is aggregated at a router and transported via an optical transponder (OT1) toward the cloud facilities over the packet-optical network infrastructure. The OT2 node represents the destination endpoint providing physical connectivity to a regional AI data center (AI-DC). The optical transport network consists of reconfigurable optical add-drop multiplexers (ROADMs) interconnected by fiber links spanning distances of several tens of kilometers. Within the AI-DC, visualization applications (*Vis. App.*) have been previously instantiated on dedicated servers to process and display the incoming sensor and data source streams.

OT1 and OT2 transponders follow a *white-box* approach, where a common open-source Network Operating System (NOS) enables seamless programmability of vendor-agnostic devices, such as coherent optical transceiver pluggable modules. A similar approach is adopted by the ROADMs within the transport network, deployed over the CTTC ADRENALINE Testbed®. Configuration and programmability of OTs and ROADMs is handled by a unified optical SDN controller instance [4, 5].

Concerning the data domain, there is a heterogeneous combination of sensors publishing data asynchronously to provide rich information for enhancing situational awareness and environmental perception. The sensors are integrated in the USEFUL platform, a sensorized vehicle that includes advanced imaging sensors (2D and 3D) and inertial sensors GPS/INS connected to a local processing unit and a Network Attached Storage (NAS) device to record all the data [6, 7]. The goal of using this variety of sensors is to provide heterogeneous information covering the largest area of the vehicle's surroundings to improve the perception of its environment. Inset of Fig.1 depicts the sensors' FOV coverage. This large amount of sensors streaming data involves a large throughput. When combined it can be up to 3.9 Gb/s.

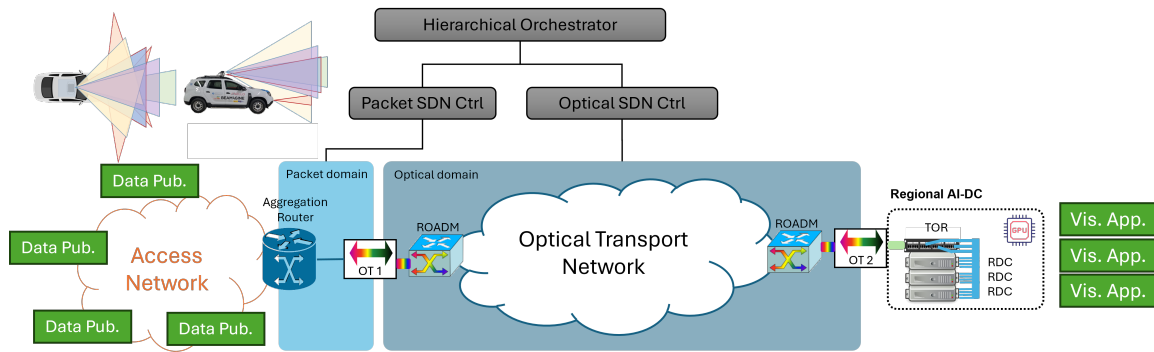


Fig. 1: Demonstration scenario.

The Robot Operating System (ROS) [8] serves as the core middleware framework to coordinate distributed sensing, processing, and communication pipelines in the service layer. ROS organizes applications into independent processing nodes working in a decoupled publish-subscribe architecture. ROS data publishers are scattered in the access network to ingest and transmit high-dimensional sensor data. These data publishers handle real-world or replayed data payloads at the exact operational frequencies and packet structures of physical hardware, ensuring that the network traffic reflects actual sensor footprints.

In turn, ROS subscribers act as data sinks or distributed processing engines located at distinct physical / logical layers of the network (e.g., in a regional AI-DC). These subscribers can perform two primary functions: real-time visualization / diagnostic monitoring; and advanced distributed perception (intermediate feature fusion). The first involves a ROS subscriber that integrates visualization tools such as ROS *rviz* to render raw multi-modal streams in real time. It can display 2D video from cameras, high-density 3D point clouds from LiDAR, target RADAR tracks, and detection labels. The second is when the ROS subscriber implements a distributed deep neural network pipeline. There, data publishers run the encoder portion of an object detection model to extract intermediate feature maps, which are compressed using a specialized 3D autoencoder [9]. The subscriber aggregates these compressed intermediate features from multiple publishers, passes them into an intermediate fusion layer [10], and processes them to synthesize a unified, occlusion-free perception map to be displayed in the ROS *rviz* interface overlaid on the raw data.

Because the applications running in this scenario are highly demanding, the capacity of OT1 (and potentially OT2) may be exceeded after deployment of several data publishers, leading to congestion. This situation is intentionally created as part of the demonstration, triggering the hierarchical orchestrator to detect the bottleneck and initiate an upgrade at the optical layer.

Another part of the demonstration focuses on a

failure emulation in the optical domain. So, restoration mechanisms are activated, allowing to observe/measure the impact on the application layer.

Novelty

The key innovations validated in this demonstration assess selected heterogeneous autonomous operations (including closed-loop control) supported by the deployed hierarchical orchestration architecture: i) dynamic network resource allocation and configuration to adapt to sensor/data source traffic variations; and ii) rapid connectivity service recovery upon failures, such as fiber link cuts. Also, this demonstration addresses the integration of ROS with the controlled packet-optical infrastructure to deploy services for next-generation mobility sensing. ROS publishers emulate users demanding high-capacity services generated by physical sensors, such as the USEFUL platform.

In fact, raw high-resolution sensor streaming requires an uplink capacity exceeding 1.6 Gb/s and up to 3.9 Gb/s, causing bottlenecks in conventional access networks. Even when advanced mid-level cooperative perception is deployed, compressing feature maps to achieve a $10\times$ reduction in communication volume the network must provide highly deterministic, high-throughput channels to preserve 3D object detection accuracy [11].

The goal is to demonstrate the effective mapping of sensing-layer service requirements onto the packet-optical network infrastructure through the hierarchical orchestration system, resulting in a synergy between sensing-as-a-service and an open, programmable packet-optical network that autonomously satisfies the rigorous real-time data fusion demands of future connected mobility.

ECOC relevance

ECOC is placing growing emphasis on the seamless integration of heterogeneous technologies across multiple network segments — spanning optical and IP/packet networks — under unified orchestration and control to deploy diverse services meeting stringent performance requirements. Our demonstration addresses this challenge by showcasing how the proposed hierarchical orchestration and its coordinated SDN controllers autonomously manage packet and optical resources

in real time, providing dynamic capacity adjustment and ensuring service continuity upon failures, to effectively support real traffic from critical applications continuously demanding high-capacity connectivity between heterogeneous data sources and regional high-performance computing infrastructures. For network operators and equipment vendors, this represents a practical step toward making packet / optical networks a key enabler for AI-driven applications, closely aligning with the ECOC community's goals of enabling multi-layer automation and zero-touch network operations.

Demo implementation

This demonstration showcases the operations of the proposed hierarchical orchestrator and its technology-specific SDN controllers to support the complete lifecycle management of connectivity services for mobility applications with stringent network requirements. Two validations are carried out: i) an increase in active data publishers triggering dynamic capacity adjustments; and ii) an optical link failure requiring immediate detection and service restoration to ensure continuity. The intended workflow follows these steps:

1) Multi-Layer Visibility & ROS Initialization:

The hierarchical orchestrator establishes end-to-end paths across packet and optical domains, enabling ROS data publishers to stream sensor data to a remote subscriber via a unified topology view.

2) Baseline QoS Monitoring: A "Digital Twin" dashboard displays real-time telemetry (e.g., packet interface utilization via gNMI, optical power levels via OpenROADM) confirming nominal performance for the mobility application.

3) Induced Traffic Congestion: Additional ROS publishers are activated to emulate peak demand, causing real-time performance degradation — increased latency, stuttering, and visual artifacts in the sensing application.

4) Autonomous Capacity Scaling: The orchestrator detects and localizes network bottlenecks, then coordinates control actions to increase transponder line-rate or spectrum allocation.

5) Post-Action Metric Validation: The GUIs confirm the sensing stream's quality recovery, validating the orchestration system's flexibility to handle varying application demands.

6) Physical Layer Fault Injection: An emulated optical link disruption (e.g., OSNR degradation) causes packet loss in the application while the optical SDN controller receives a gNMI notification.

7) Optical Path Restoration: The optical SDN controller autonomously computes and configures the path to restore the affected services.

8) Service Continuity Verification: Attendees witness the automated recovery of the sensing stream, validating the resilience of the disaggregated transport layer for critical mobility services.

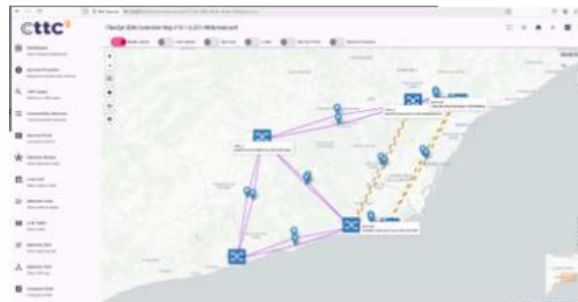
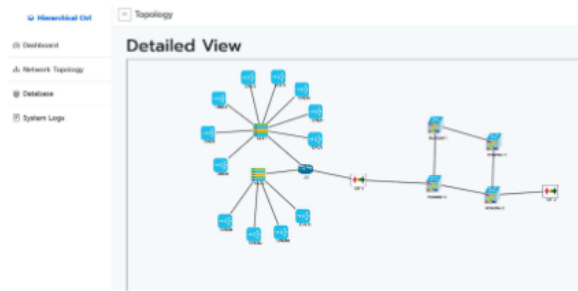


Fig. 2: Sample GUI screenshots of the hierarchical orchestrator and the optical controller.

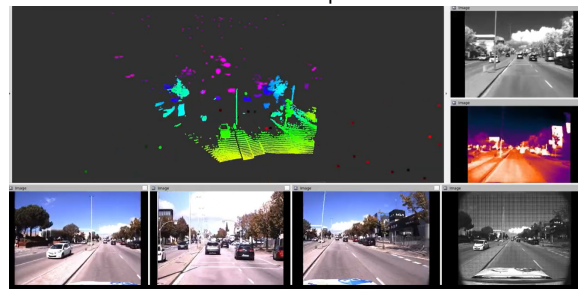


Fig. 3: Screenshot of the application deployed at the regional AI-DC when visualizing data from the data publishers.

ECOC attendees will follow both congestion- and failure-recovery workflows — covering detection, path computation, and device reconfiguration — while observing the impact on the end-user application (artifacts and stuttering prior to recovery), and interacting directly with the GUIs shown in Fig.2 and 3. Fig. 2 shows the dashboards of the hierarchical orchestrator and the optical network controller. Fig. 3 shows the data processing and visualization dashboard, illustrating how network status affects the application. Through these GUIs, attendees can monitor messages between network control systems, resource occupancy, and the physical/logical topologies at each demo step.

We will emulate the data stream of the USEFUL car with recorded real data by deploying dedicated ROS publishers in the physical layer. These publishers will feed the data visualization subscribers deployed in the regional AI-DC with the exact same data (sensor messages, timing and topic structure) as the real vehicle sensors. Thus, attendees will experience a realistic end-to-end traffic demonstration and data visualization. In fact, a plurality of publishers will be simultaneously deployed, emulating multiple vehicles collaborating with high-end data to the network.

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